

Haider Ali

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ROBOTICS SOFTWARE & EMBEDDED SYSTEMS ENGINEER

Mechatronics Engineer specialising in **robotic autonomy, SLAM, and sensor fusion**, with strong hands-on experience in **Modern C++, Python, ROS-based systems, and embedded Linux**. Proven ability to design, integrate, and deploy **field-ready robotic and sensing platforms**, spanning real-time data pipelines, custom drivers, and system-level integration. Strong interest in translating research concepts into reliable robotic systems for real-world deployment.

ROBOTICS SYSTEMS & TECHNICAL OWNERSHIP

- Owned end-to-end software and system integration for autonomous and sensing platforms, from low-level drivers to operator-facing dashboards.
- Led integration of multi-sensor robotic systems, coordinating software, hardware, and deployment constraints.
- Supported field deployment, validation, and reliability testing of robotic and embedded systems in outdoor and operational environments.

EXPERIENCE

Embedded Engineer	Robionix Technologies	Sep 2024 – Present
• Led the design and system integration of a LoRa-based autonomous irrigation platform, implementing real-time scheduling and mesh networking on embedded Linux to achieve sub-second command response in field conditions. • Designed and integrated a multi-sensor robotic sensing platform (GPS, camera, flame, smoke, motion), developing custom Linux device drivers to ensure deterministic data flow and real-time event handling. • Implemented a fault-tolerant LoRa fallback communication system, enabling continuous operation and emergency reporting without internet connectivity. • Planned and executed environmental and climatic validation testing for deployed hardware, improving robustness and long-term reliability. • Integrated embedded platforms with centralized monitoring dashboards, enabling real-time system visibility and operator decision support.		
Design Engineer	Orbon Technologies	Jun 2023 – Jul 2024
• Developed an assistive mobile robotic system featuring Urdu voice-controlled navigation using TensorFlow on Raspberry Pi, enabling accessible human-robot interaction. • Designed a VFD-based motor control system with feedback-driven thermal modulation, reducing inrush current and extending motor lifespan by approximately 20%. • Engineered an RFID-based real-time race tracking system, integrating embedded Linux backends with web dashboards and SMD displays for live event monitoring.		
Mechatronics Intern	NCRA, NUST EME	Jun 2022 – Aug 2022
• Developed an autonomous outdoor UGV using RTK-GPS and compass-based sensor fusion, achieving sub-centimeter localization accuracy for large-scale navigation. • Implemented a Linux kernel module for low-latency sensor acquisition, enabling real-time data delivery suitable for SLAM and closed-loop robotic control.		

EDUCATION

BSc Mechatronics Engineering, Air University, Islamabad

CGPA: 3.03 / 4.00

2020 – 2024

TECHNICAL SKILLS

Robotics & Autonomy: SLAM, Sensor Fusion, Localization, RTK-GPS, Extended Kalman Filter, Path Planning
Programming: Modern C++ (C++14+), Python, Embedded C, Bash
Robotics Middleware: ROS (working knowledge, system integration)
Systems & Software: Embedded Linux, Real-Time Systems, System Integration, IPC (MQ, Shared Memory)
Build & Tooling: CMake, Makefile

SELECTED ROBOTICS PROJECTS

Autonomous Outdoor Mobile Robot (RTK-GPS)

- Designed and integrated an autonomous UGV using RTK-GPS and compass-based sensor fusion.
- Implemented grid-based path planning and mapping to support large-scale outdoor navigation and SLAM workflows.

RFID-Based Marathon Tracking System

- Developed an embedded Linux-based RFID tracking system with real-time duplicate-scan prevention and live leaderboard generation.
- Integrated on-site SMD displays and web dashboards to support live race operations.

Self-Balancing Two-Wheel Robot

- Implemented gyroscope-based state estimation and PID control, achieving stable real-time balance on dynamic surfaces.

CERTIFICATIONS

ISTEC Vibration Training Category 1+ (ISO 18436-2)

Valid: 2024 – 2029